What separates the 149 fills that reached 1R from the 181 that didn't

Run `RUN_2026_07_28_22_45_40_17630` (12 months, Jul 2025 – Jul 2026). 330 filled trades, 326 with a parseable `fill_time`. Split into a first half (to 9 Jan 2026, n=158) and a second half (n=168) as a temporal holdout.

Base rate: 45.2% of fills reach 1R. First half 42.4%, second half 47.6%. The difference between halves is not significant, so the two halves are a fair test of each other.

Method

Every field in `merged_setups.csv` that could only be known after the trade was over was excluded before testing. That removed the whole MFE family, every `sim`/counterfactual/`would_trade_have_*`/`did_reach_*` field, all `reached_*` fields other than the target itself, `was_stop_hit`, `minutes_until_max_rr`, `reversal_from_max_rr_pips`, `pct_of_target_completed_before_reversal`, all `m4_*` money and exit fields, all account equity and balance fields, all `next_setup_*` fields, and all `*_after_entry` counters. What remained was 241 testable comparisons over fields that exist at or before the moment the order fills — the `entry_known_*` family, block geometry, sweep geometry, anchor geometry, liquidity-map fields, session and regime labels, spread and slippage at fill, and the funnel timing fields.

Continuous fields were tested with Mann-Whitney, categorical levels with Fisher's exact, and every hit was then re-tested independently in each half.

Result

Of 241 comparisons, 11 came back at $p < 0.05$. Pure chance predicts 12.1. The scan found less than random noise would have produced.

The three that looked strongest all fail on inspection rather than on statistics:

event_count_for_setup (p = 3.7e-17) is not a pre-trade field despite the name. A setup that runs longer logs more events, and a trade that reaches 1R by definition ran longer. It is the outcome wearing a different label.

lots_sent / lots_raw (p = 3.3e-05) is entirely a symbol confound. Median lots by symbol: XAGUSD 0.05, XAUUSD 0.07, EURUSD 0.49, GBPUSD 0.63, GBPAUD 0.86, USTECH100M 1.00, USDCAD 1.12, AUDJPY 1.19, EURAUD 1.19, CHFJPY 1.22, GBPJPY 1.25, USDJPY 1.52. The metals dominate the low-lot group and the JPY crosses the high-lot group, so "small lots reach 1R more often" is just "silver reaches 1R more often than the yen crosses", which is a per-symbol observation with n = 67 and n = 30 behind it, not a tradeable rule.

nearest_liquidity_below_touch_count and nearest_liquidity_below_score are the only pair that point the same way in both halves — fills that reached 1R sat below liquidity with fewer touches (median 20 vs 22) and a lower score (305 vs 330). But the effect is 2 touches on a 20-touch base, it is only significant in the second half (p = 0.017 vs p = 0.09 in the first), and inside individual symbols it survives in exactly one of five: GBPAUD p = 0.027, XAGUSD p = 0.17, AUDJPY p = 0.11, CHFJPY p = 0.29, GBPJPY p = 0.96. One symbol out of five at p = 0.03 across five tests is what chance looks like.

The structural fields, for completeness

None of these separates the two groups, and the reach-1R rate is flat across almost every cut:

By timeframe — M1 48% (n=223), M5 44% (n=82), M15 24% (n=21). M15 is the only genuinely low cell and it holds in both halves (20% then 27%), but 21 trades is too few to act on.

By session — OffSession 49%, NewYork 46%, London 45%, Asia 38%. Asia is the weakest in both halves but only by a few points on 56 trades.

By direction — sell 46% (n=243), buy 42% (n=83). No skew in reach-1R despite the 4:1 short bias in setups.

By block type — single_candle 46% (n=193), hybrid_2_candle 43% (n=113), area 50% (n=16). Flat.

By symbol — GBPUSD 58%, EURAUD 53%, AUDJPY 52%, GBPAUD 48%, XAGUSD 48%, GBPJPY 40%, CHFJPY 37%, EURUSD 29%. And these do not hold across halves: GBPJPY goes 29% then 57%, CHFJPY 30% then 50%, EURUSD 50% then 20%. Per-symbol reach-1R is not stable.

By volatility regime — high 54%, low 46%, normal 33%. Does not hold: "high" is 65% in the first half and 50% in the second, "normal" is 42% then 25%.

By trend regime — downtrend 50%, range 42%, uptrend 38%. Weak and unstable.

By day of week — 41% to 50%, no pattern.

`entry_model` is `liquidity_block_retest` for all 326. `anchor_tf` is M15 for all 326. `market_regime_label` is `unknown` for all 326. These three fields carry no information at all in the filled population.

Conclusion

Nothing recorded in the data at the moment of entry distinguishes a trade that goes 1R in favour from one that goes straight to the stop. The signal is not hiding in a field that needs a better threshold; there is no signal in any of the 241 fields tested.

This is a stronger result than it first looks, and it narrows the search rather than ending it. It means the difference is not something the EA already measures. Any fix has to come from something that is not currently in the feature set — a condition the system does not yet look at, at a point in the sequence it does not yet check. Threshold-tuning on existing gates (`min_rr_required`, quality score, block height limits, spread filters, session filters, symbol removal) cannot work, because none of those fields separates the two groups even in-sample, let alone out of it.

The one thing worth flagging as a possible structural finding rather than noise: the target-distance picture. 281 of 326 fills sit in the `extreme` target-distance bucket and 304 of 326 in the `5R_plus` quality bucket. Median `distance_entry_to_target_pips` is around 1,500 pips. The system is almost never taking a trade with a nearby target, so "target distance" has effectively no variance to analyse — it is a constant, not a variable. That is not evidence that closer targets would help (that question is closed), but it does mean the 12-month dataset contains no counterexample to test it against either way.

Related: debug-flag safety verification

Verified this session, in response to the question of whether disabling `DEBUG_M2`, `ENABLE_M2_VERBOSE` or `ENABLE_M3_VERBOSE` breaks anything.

All three helper functions called inside debug guards were read in full:

P(key, msg) at line 261 — keeps a static array of 2,048 key/hash pairs so it only prints when a message changes. The static arrays are private to the function and read by nothing else. Print-only.

PrintThrottled(key, msg, everySec) at line 5026 — same shape, a 512-slot static ring buffer of keys and timestamps, private to the function. Print-only.

M3_Skip(sym, tf, bz, reason) at line 9726 — builds a format string, calls `P()`, and returns the string. It has exactly two call sites, lines 10004 and 10030, and both discard the return value. It reads `bz.setup_status` and `bz.block_id` but writes nothing. Print-only.

A separate scan for assignments to struct members or globals inside any block guarded by the three flags returned nothing.

All 20 `FileWrite` / `FileWriteString` calls in the file were located and none has any of the three flags within 40 lines above it. Data emission to the NDJSON and CSV files is entirely independent of the debug flags.

Conclusion: all three flags are safe to switch off in the Strategy Tester Inputs tab. No recompile is needed and no data is lost.